



UNIVERSITY OF LIMPOPO

FACULTY OF SCIENCE AND AGRICULTURE

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

DEGREE AND DIPLOMA PROJECT 1

DUE DATE: 04 MAY 2026

MODULE: SCO031

PAPER: P

(INTRODCUTION DATABASES)

DURATION: 15 DAYS

MARKS: 100

INTERNAL EXAMINER: DR J TLOUYAMMA

THIS PAPER CONSISTS OF (10) PAGES INCLUDING COVER PAGE

INSTRUCTIONS:

1. Complete all tasks in **PART A – PART F**
2. For deliverables, please refer to **item 3 on page 7**
3. In addition to deliverables, include the following:
 - 3.1. **Attendance registers for every group meeting**
 - 3.2. **Group members' declaration form**
4. Submission will be through **blackboard Dropbox**.
5. All documents to be packaged into zipped folder named by **project name & Group no**
6. These submissions are due on the **4th of May 2026 at 12:00 mid-night**
7. **No late submissions will be accepted**

ENTERPRISE DATABASE MANAGEMENT & MONITORING SYSTEM

1. Project Overview

You are appointed as a **Database Administrator (DBA)** for a company using the AdventureWorks 2022 database. Your responsibility is to design, implement, and manage a **secure, efficient, and monitored database system**.

You must demonstrate your ability to:

- Automate database operations
- Secure access using roles
- Monitor and improve performance
- Track changes using triggers
- Send automated alerts to the DBA

2. Project Requirements

You are required to complete **ALL** the following tasks:

PART A: Stored Procedures (Database Automation)

Objective

Create stored procedures to manage core database operations.

◆ Tasks

1. Data Management Procedures

Create stored procedures for:

- Adding a new customer
- Updating product prices
- Deleting or archiving inactive customers

Requirements:

- Use input parameters
- Validate input data
- Use transactions (BEGIN TRANSACTION)
- Implement error handling (TRY...CATCH)

2. Reporting Procedures

Create stored procedures that generate:

- Monthly sales report

- Top 10 best-selling products
- Employee performance summary

Requirements:

- Accept parameters (e.g., date range)
- Use joins and aggregations
- Return meaningful result sets

PART B: User Roles & Security

Objective

Implement **role-based access control**.

◆ Tasks

1. Create Roles

Create the following roles:

- SalesRole
- HRRole
- DBA_Role

2. Assign Permissions

Role	Permissions
SalesRole	Read/write sales data
HRRole	Access employee data
DBA_Role	Full control

Use:

- GRANT
- DENY
- REVOKE

3. Create Users

- Create at least 3 users
- Assign each user to a role
- All members of the group should have usernames associated with any of the roles above

4. Testing

- Demonstrate restricted access
- Show that users cannot access unauthorized data

PART C: Performance Optimization

Objective

Improve database performance.

◆ Tasks**1. Indexing**

- Identify or create slow queries
- Create appropriate indexes
- Compare performance (before vs after)

2. Query Optimization

- Rewrite inefficient queries from slow executing queries above
- Use execution plans

3. Monitoring Procedures

Create stored procedures to:

- Detect long-running queries
- Check database size
- Monitor index fragmentation

PART D: Backup & Recovery**Objective**

Ensure data safety.

◆ Tasks

- Use SQL query to create:
 - Full backup
 - Differential backup
 - Transaction log backup
- Demonstrate:
 - Restore process
- Provide:
 - Backup schedule (documentation)

PART E: Triggers, Logging & Email Alerts**Objective**

Track and respond to database changes automatically.

◆ Task 1: Create Audit Table

Create a table named AuditTable to log all changes:

- Table name
- Action type (*note: this column contents should either be INSERT, UPDATE, or DELETE depending on the nature of operation performed*)
- User
- Date/time
- Description

The table should keep track of insertions, deletions and updates made to two tables (Customer, Product) and be updated by the trigger.

◆ Task 2: Create Triggers

Create triggers for:

- INSERT
- UPDATE
- DELETE

Apply triggers on TWO tables (Customer, Product).

◆ Task 3: Logging

Each trigger must:

- Record the action in the AuditTable
- Capture relevant details under the Description column (e.g., ProductID, CustomerID). For example: “The record that was deleted by {put username that deleted the record here} belong to customer with CustomerId: {Put customer Id here}”

◆ Task 4: Email Alerts

Configure **SQL Server Database Mail** and:

- Send email notifications when:
 - A product price changes
 - A record is deleted
 - A critical update occurs

◆ Task 5: Conditional Alerts (Advanced)

Only send alerts when:

- Price change > 10%
- Sensitive data is modified

 **Important:**

- Avoid heavy processing in triggers
- Ensure system performance is not degraded

PART F: Maintenance & Server Health

Objective

Maintain database health.

◆ Tasks

- Create maintenance plan including:
 - Index rebuild/reorganize
 - Database integrity check (DBCC CHECKDB)
 - Scheduled jobs (SQL Server Agent)

3. Deliverables

You must submit:

1. SQL Scripts

Include:

- Stored procedures
- Role and user creation scripts
- Index scripts
- Backup scripts
- Triggers

2. Documentation Report (Compulsory)

Your report must include:

a) System Design

- Explanation of all components

b) Security Model

- Roles and permissions

c) Performance Improvements

- Before vs after comparison

d) Trigger & Alert Design

- How logging and email system works

e) Backup Strategy

- Schedule and recovery plan

3. Demonstration (Practical)

You must demonstrate:

- Stored procedures working
- Role-based access restrictions
- Trigger logging in action
- Email alerts being sent
- Performance improvements

4. Skills Assessed

- SQL programming
- Database security
- Performance tuning
- Automation (triggers & procedures)
- Backup & recovery
- Problem-solving

5. Submission Guidelines

- Submit all files in a **single ZIP folder**
- Include:
 - SQL scripts
 - Report (PDF/Word)
- Clearly label your files

SCOA031 PROJECT MARKING RUBRIC

Marking Rubric (Total: 100 Marks)

PART A: Stored Procedures (20 Marks)

Criteria	Marks	Description
Data management procedures	6	Procedures correctly perform INSERT, UPDATE, DELETE with parameters
Use of transactions	4	Proper use of BEGIN TRANSACTION, COMMIT, ROLLBACK
Error handling	4	Effective use of TRY...CATCH
Reporting procedures	4	Accurate queries with joins, aggregations, parameters
Code quality & structure	2	Clean, readable, well-named procedures

PART B: User Roles & Security (10 Marks)

Criteria	Marks	Description
Role creation	4	Correct creation of all required roles
Permission assignment	5	Proper use of GRANT, DENY, REVOKE
User creation & mapping	3	Users correctly assigned to roles
Security testing	3	Clear demonstration of restricted access

PART C: Performance Optimization (10 Marks)

Criteria	Marks	Description
Index creation	5	Appropriate indexes created for performance improvement
Query optimization	4	Inefficient queries improved
Performance comparison	4	Evidence of before vs after improvement
Monitoring procedures	2	Procedures to monitor performance implemented

PART D: Backup & Recovery (10 Marks)

Criteria	Marks	Description
Backup implementation	4	Full, differential, and log backups created
Restore process	3	Successful demonstration of restore
Backup strategy documentation	3	Clear and realistic schedule

PART E: Triggers, Logging & Email Alerts (20 Marks)

Criteria	Marks	Description
Audit table design	3	Well-structured logging table
Trigger implementation	6	Correct INSERT, UPDATE, DELETE triggers
Logging accuracy	4	Logs contain meaningful and correct information
Email alerts (Database Mail)	4	Alerts sent successfully using SQL Server Database Mail
Conditional logic	3	Alerts triggered only under defined conditions

PART F: Maintenance & Server Health (10 Marks)

Criteria	Marks	Description
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Maintenance plan	4	Includes index maintenance and integrity checks
Use of DBCC CHECKDB	2	Proper implementation
Automation (SQL Server Agent jobs)	4	Tasks scheduled and explained

DOCUMENTATION REPORT (10 Marks)

Criteria	Marks	Description
System explanation	3	Clear explanation of all components
Security model	2	Roles and permissions clearly explained
Performance discussion	2	Evidence of optimization
Backup & recovery explanation	2	Clear and logical
Clarity & presentation	1	Well-structured, professional

DEMONSTRATION / PRACTICAL (10 Marks)

Criteria	Marks	Description
Stored procedure execution	2	Demonstrates working procedures
Role-based access	2	Shows restricted access
Trigger logging	2	Logs generated correctly
Email alerts	2	Emails triggered successfully
Performance improvements	2	Demonstrated clearly

FINAL MARK SUMMARY

Section	Marks
Part A: Stored Procedures	20
Part B: Security	10
Part C: Performance	10
Part D: Backup	10
Part E: Triggers & Alerts	20
Part F: Maintenance	10
Documentation	10
Demonstration	10
Total	100

Performance Level Guide

Level	Percentage	Interpretation
Excellent	75–100%	Industry-ready DBA skills
Good	60–74%	Solid understanding
Satisfactory	50–59%	Basic competency
Poor	<50%	Needs improvement